GUARDIANSIM

GuardianSim: Counterfactual Safety Certification for Robot Manipulation on AMD Radeon GPUs

Track 3 - Physical AI Challenge
AMD Radeon Cloud / ROCm / Genesis simulation

SUBMISSION EDITION - VERIFIED SIMULATION EVIDENCE

Project repository
github.com/nvm-star-max/GuardianSim

Team attribution
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Counterfactual safety certification for robot manipulation on AMD Radeon GPUs

Abstract

Robot manipulation policies are usually optimized to complete a task, but a nominally successful grasp can still pass too close to nearby objects, contact clutter, or become unstable when scene geometry changes. GuardianSim is a safety-oriented decision layer for simulated Franka Panda fruit picking. Given a nominal grasp, it generates counterfactual alternatives, restores every alternative to the same Genesis scene snapshot, measures physical rollout outcomes on an AMD Radeon GPU, and selects an action only when it satisfies frozen reachability, stability, clearance, and repeatability requirements. If no candidate passes every hard gate, the system explicitly stops.

In the primary frozen Gate 3.2 benchmark, GuardianSim achieved repeatable safe completion in 30 of 30 declared clutter scenarios, compared with 18 of 30 for the nominal baseline. Across three independent physical executions per scenario, safe executions improved from 58/90 to 90/90, while sampled clutter contacts decreased from 30 to zero. Mean sampled clutter clearance increased from 0.023191 m to 0.046003 m. These are Genesis simulation results on AMD Radeon Cloud, not physical-robot deployment claims.

The completed Safety Swarm V2 run tested the decision layer at a larger candidate-by-uncertainty scale: 18 actions across 256 declared uncertainty worlds each, for 4,608 measured pairs. Five actions passed all 256 worlds. The frozen ranking selected one action with 256/256 safe outcomes, zero sampled clutter contacts, 66.249 mm worst-case sampled clearance, and 0.907 minimum stability. This is an engineering stress-test population, not 4,608 independent robot trials.

A separate Radeon throughput run measured the same GPU backend at 1, 16, 64, and 256 parallel Genesis worlds. The 256-world point reached 35,166.1 environment-steps/s, 228.16 times the single-world throughput, at 89.1% parallel efficiency. This is compute evidence, not an increase in the formal safety sample count.

1. Target application and motivation

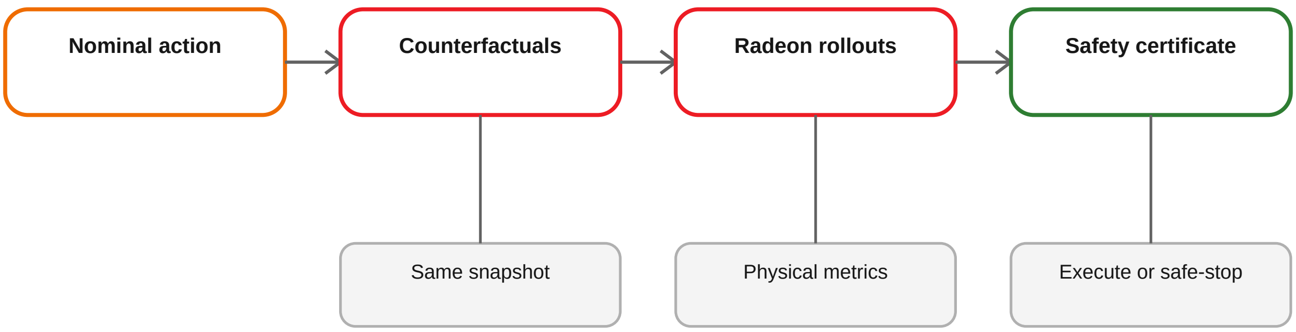
The target application is safety-aware action selection for robot manipulation in clutter. The retained reference task uses a Franka Panda arm to pick a banana, lemon, or plum from a table and move it toward a bowl. A nominal scripted policy can complete many layouts, but its fixed grasp orientation and approach geometry do not explicitly reason about nearby clutter.

GuardianSim addresses a narrower and auditable question:

> *Before executing a policy-proposed grasp, can a simulator evaluate bounded alternatives from the same physical state, reject actions that violate safety margins, and either execute a safer eligible action or stop?*

The project focuses on the decision boundary rather than claiming a new end-to-end robot foundation model. This makes every action choice inspectable and every published metric traceable to a preserved report.

2. System architecture



The implementation is split into four layers:

1. **Simulator-independent decision core.** Candidate models, scoring, recovery logic, serialization, and benchmark validators do not depend on Genesis runtime objects.
2. **Snapshot-safe simulation adapter.** A captured episode state contains robot joints, object poses, and a deterministic fingerprint. Every candidate rollout restores this state before evaluation.
3. **Physical measurement layer.** Rollouts produce reachability, grasp alignment, retained-lift stability, path length, uncertainty, and minimum sampled clutter clearance. Support-surface contact is diagnosed separately from non-target clutter overlap.
4. **Safety-first selector.** Hard gates determine eligibility before utility ranking. The selector never falls back to an unsafe nominal action.

3. Method

3.1 Counterfactual action space

The initial candidate matrix varied grasp yaw and lateral gripper offsets. The final Gate 3.2 action space was expanded after the Gate 3.1 negative result. It contains 18 obstacle-aware candidates:

- yaw angles from -90 degrees to +90 degrees in 22.5-degree increments;
- the nominal target position;
- a 0.025 m retreat direction away from primary clutter;
- a raised 0.14 m approach for retreating candidates.

The nominal action remains represented explicitly, so GuardianSim can retain it when it is already safe.

3.2 Snapshot identity

Comparing candidates from different physical states would invalidate the counterfactual claim. GuardianSim therefore:

- captures one settled episode snapshot;
- fingerprints its robot and object state;
- restores the same snapshot before each candidate rollout;
- validates the fingerprint when resuming a report.

During development, strict resume validation correctly rejected two attempts to append a smoke report from a newly initialized Genesis process whose base snapshot fingerprint differed. Those rejected logs are preserved as audit evidence.

3.3 Physical measurements

For each candidate, GuardianSim measures:

- inverse-kinematics reachability;
- grasp-axis alignment;
- retained object lift and stability;
- sampled path length;
- perception uncertainty;
- minimum AABB separation between distal arm links and named non-target clutter.

A diagnostic identifies the responsible sample, robot link, obstacle, overlap state, and overlap depth. Table support contact is recorded separately and does not determine clutter collision risk.

3.4 Safety eligibility and repeatability

Gate 3.2 freezes all thresholds before the formal run. A candidate is eligible only if its conservative aggregate passes every reachability, stability, and minimum-clearance gate. Candidate confirmation uses repeated rollouts; formal execution then independently executes each strategy three times. An episode is a repeatable safe completion only when all three executions pass.

The selector returns one of four explainable decision types:

- retain an eligible nominal action;
- use a higher-margin alternative;
- replace an unsafe nominal action;
- safe-stop because no action is eligible.

3.5 Post-condition monitoring

Independent execution records whether the gripper closed, object lift was retained, clutter was contacted, the clearance threshold was violated, or the lift became unstable. Failure diagnosis maps these observations to bounded recovery actions. The benchmark does not count a visually plausible but unsafe execution as a success.

4. Training and evaluation data

GuardianSim's primary results do not require a newly trained policy or an external training dataset. The nominal action comes from the retained scripted Franka fruit-picking reference pipeline. The contribution is the counterfactual safety layer and its physical evaluation.

The frozen Gate 3.2 evaluation matrix contains 30 procedurally declared scenarios. The project does not use them as a statistical training holdout because the evaluated nominal policy is scripted rather than learned:

- three target objects: banana, lemon, and plum;
- lateral and radial clutter configurations;
- five deterministic seeds per object/configuration cell;
- three independent physical executions for each strategy and scenario.

The scenario order, thresholds, protocol hash, and matrix hash were frozen before the formal outcomes were inspected:

- protocol SHA-256: `8f23247001e05f39817225ed13f028321fbb9b9c694aaacd5b987fe61ee1fb3c`;
- matrix SHA-256: `69f87994b87f2def788cd944ad75210cdeddeafcaa3d0a3844fef04efca9cb03`.

Gate 3.3 adds engineering breadth tests for pose shift and changed gap/bearing. Its completed 12-scenario prefix is reported separately and is not combined with the primary 30-scenario performance claim.

Safety Swarm V2 uses a separate frozen uncertainty matrix for action screening. Each of the 18 Gate 3.2 action geometries is evaluated against the same 256 deterministic worlds. Its unit of analysis is a candidate-world pair, so its 4,608 outcomes are never added to the Gate 3.2 scenario or execution counts.

5. AMD Radeon GPU use

The formal benchmark and evaluator smoke ran on AMD Radeon Cloud using the Genesis `gs.amdgpu` backend.

Recorded formal environment:

- AMD Radeon Cloud Blank OpenCode Workspace;
- one AMD Radeon device, PCI model `0x744b`, GFX architecture `gfx1100`;
- approximately 47.98 GB device memory reported by Genesis;
- Python 3.12.3;
- PyTorch `2.9.1+gitff65f5b`;
- HIP `7.2.53211-e1a6bc5663`;
- Genesis 1.2.3.

The GPU accelerates Genesis scene stepping and the repeated physical counterfactual rollouts. Mean Gate 3.2 planning wall time was 264.95 seconds per scenario for the frozen candidate and confirmation protocol. Mean independent execution time was 9.08 seconds for the baseline and 8.94 seconds for GuardianSim.

The repository includes exact ROCm wheel installation, a dependency lock, an environment-manifest collector, a GPU-required preflight, a bounded three-candidate Genesis smoke, and a complete-source ROCm Dockerfile.

5.1 Measured parallel-physics scale

The scale run used one fixed headless Franka scene, 100 warmup steps, and 1,000 timed steps per batch size. Scene construction and JIT warmup were excluded from the timed interval.

Parallel worlds	Environment-steps/s	Speedup vs. 1 world	Parallel efficiency
1	154.1	1.00x	100.0%
16	2,383.7	15.47x	96.7%
64	9,354.3	60.69x	94.8%
256	35,166.1	228.16x	89.1%

All four points reached 96% peak GPU utilization. At 256 worlds, mean GPU utilization was 85.5% and peak VRAM use was 1.34 GiB. The complete timed workload contained 337,000 environment steps.

5.2 Safety Swarm V2 formal decision-scale run

The final run evaluated all 18 candidates against all 256 frozen uncertainty worlds: 4,608 candidate-world pairs, executed as deterministic candidate-major chunks on one Radeon GPU. It advanced 2,299,392 Genesis environment steps in 226.676 seconds, or 10,143.979 environment-steps/s and 20.329 candidate-world pairs/s.

Radeon telemetry recorded 73.406% mean and 97% peak GPU utilization over 588 samples, with 1.381 GiB maximum VRAM use and no sampling errors. Across all alternatives, 2,614 pairs were safe, 270 recorded sampled clutter contact, 1,691 failed the stability gate, and 33 failed the clearance gate.

Five candidates passed every one of their 256 worlds. Frozen ranking selected `yaw_-45.0_retreat_+0.000_approach_+0.140`, which recorded 256/256 safe worlds, zero sampled clutter contacts, 66.249 mm worst-case sampled clearance, 66.304 mm fifth-percentile sampled clearance, and 0.907 minimum stability.

An earlier 54-world run remains a bounded engineering smoke, but it is no longer the project's main scale result. Neither the 4,608 formal pairs nor the 54 smoke futures are independent physical-robot trials.

6. Results

6.1 Primary Gate 3.2 result

Metric	Nominal baseline	GuardianSim	Difference
Repeatable safe completion	18/30 (60%)	30/30 (100%)	+40.00 percentage points
Independent safe executions	58/90	90/90	+32 executions
Clutter-contact executions	30	0	-30
Mean sampled clutter clearance	0.023191 m	0.046003 m	+98.36%
Mean retained-lift stability	0.892762	0.905099	+0.012337

GuardianSim decisions comprised 11 higher-margin alternatives, 10 replacements of unsafe nominal actions, and nine eligible nominal fallbacks. No Gate 3.2 scenario required a safe stop.

6.2 Formal decision-scale result

The Safety Swarm V2 result is summarized as `4,608 -> 5 -> 1`: 4,608 measured candidate-world pairs, five candidates passing all 256 worlds, and one action chosen by a ranking rule frozen before the result was inspected. The complete report, validator receipt, environment, logs, source identity, and checksums are preserved under `docs/evidence/safety-swarm-v2-formal-2026-07-30`.

The full population result is retained, including rejected alternatives. It shows that the selected action was not an isolated successful rollout and that unsafe candidate-world outcomes were not hidden from the decision record.

6.3 Negative evidence and iteration

Gate 3.1 is intentionally retained. It showed that increasing average clearance alone did not improve safe-completion rate. This negative result motivated three Gate 3.2 changes:

- expanded obstacle-aware action geometry;
- an explicit rule forbidding fallback to an unsafe nominal action;
- three independent physical executions for repeatability.

6.4 Breadth evidence and limitation

In the independent 12-scenario Gate 3.3 prefix, GuardianSim executed ten actions safely and safe-stopped twice, with zero clutter contacts or clearance-violating executions. The nominal baseline completed seven safely, made four clutter contacts, and had one clearance violation.

The two stops occurred in lateral lemon and plum gap/bearing cases where no candidate satisfied all frozen hard gates. This is correct fail-safe behavior, but it exposes a real action-space coverage limitation. GuardianSim does not claim universal completion under arbitrary geometry.

7. What GuardianSim adds

1. **It sits between a task policy and execution.** The policy still proposes the nominal action; GuardianSim checks bounded alternatives before the robot moves.
2. **Candidate comparisons start from one state.** Every rollout restores the same fingerprinted snapshot. An incompatible cross-process resume is rejected instead of silently mixing evidence.
3. **Safety gates run before utility ranking.** A candidate cannot trade a failed clearance or stability requirement for a higher score. If no action passes, the result is a visible stop.
4. **The published result requires repeatability.** A scenario counts only when all three independent executions are safe.
5. **The report explains each decision.** It records the measured clearance, stability, responsible link and obstacle, decision type, protocol identity, logs, and checksums.
6. **The Radeon path is measured at two useful grains.** The raw scale suite isolates 1-to-256-world physics throughput. The formal Safety Swarm V2 run shows how Radeon batching supports a complete 18-by-256 action decision. Neither result is added to the 30-scenario safety sample count.

8. Reproducibility and deliverables

Primary deliverables:

- source code and bundled simulation assets;
- `uv.lock` and exact Radeon wheel installer;
- complete-source ROCm Dockerfile;
- machine-readable environment capture;
- one-command source/GPU preflight;
- bounded real Genesis counterfactual smoke;
- immutable Gate 3.2 schema-5 report and validator;
- Gate 3.3 breadth evidence;

- strict Radeon scale report;
- complete Safety Swarm V2 4,608-pair report, validator, logs, and checksums;
- annotated comparison demo and interactive showcase;
- technical report and 3-5 minute video.

The evaluator path is documented in the repository root `REPRODUCIBILITY.md`. The strict validator must accept the complete 30-episode report and its frozen protocol hash before any primary metric is used.

The 4 minute 41.5 second English submission video binds the accepted Seed 411 replay, aggregate Gate 3.2 result, Gate 3.3 limitation evidence, and simulation-only claim boundary. Its SHA-256 is recorded in the package checksum manifest.

An 80-second supplementary Radeon preview shows the scale curve and earlier bounded candidate funnel; it does not replace the complete workflow video. The public showcase is the judge-facing view of the later 4,608-pair formal run and links to its immutable report. Its file identity is also recorded in the package checksum manifest.

9. Limitations and responsible claims

- All evidence comes from Genesis simulation.
- No physical robot was tested.
- Axis-aligned sampled clearance is an engineering proxy, not a formal continuous-time collision proof.
- The evaluated object set is limited to three YCB fruits and declared clutter configurations.
- Planning is not yet real-time.
- Safe stopping protects against unsupported geometry but reduces task completion.
- The reported 30-scenario result applies only to the frozen Gate 3.2 matrix.
- The scale benchmark measures steady-state Genesis stepping after warmup; it excludes scene construction and does not measure policy-training throughput.
- The 4,608 Safety Swarm V2 pairs are a candidate-by-uncertainty engineering population, not 4,608 independent robot trials.

Future work should expand continuous grasp geometry, use richer collision distance models, calibrate perception uncertainty from camera observations, reduce planning latency through parallel rollouts, and validate on physical hardware.

10. Team contributions

Member	Contribution
Solo developer - GitHub `@nvm-star-max`	Project direction, system design, implementation, Radeon Cloud experiments, evidence preservation, documentation, and demo production.

AI-assisted development tools were used for implementation and documentation support. The submitting team remains responsible for technical validation, claims, licenses, and competition-rule compliance.

References

1. GuardianSim source and evidence (https://github.com/nvm-star-max/GuardianSim)
2. Genesis Embodied AI (https://github.com/Genesis-Embodied-AI/Genesis)
3. LeRobot (https://github.com/huggingface/lerobot)
4. ROCm documentation (https://rocm.docs.amd.com/)
5. Upstream Franka fruit-picking reference (https://github.com/wangxunx/franka_fruit_pick_demo)